

Unit –II

Theory of Utility and consumer's equilibrium

Meaning and Concept of Utility

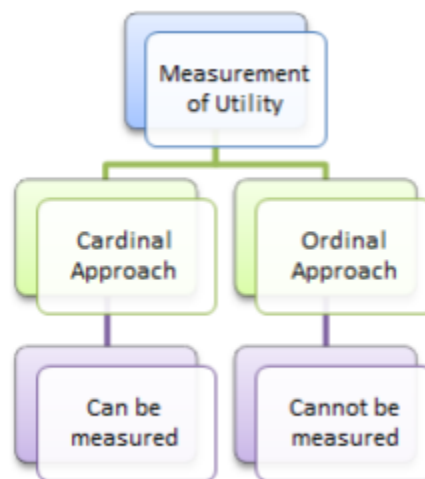
Utility Definition – It is a measure of satisfaction an individual gets from the consumption of the commodities. In other words, it is a measurement of usefulness that a consumer obtains from any good. A utility is a measure of how much one enjoys a movie, favourite food, or other goods. It varies with the amount of desire.

One can conclude the following conclusions

1. A Utility of a good differs from one consumer to another.
2. It keeps on changing for the same consumer due to change in the amount of desires.
3. It should not be equated with its usefulness.

Measurement of Utility

Measurement of a utility helps in analyzing the demand behaviour of a customer. It is measured in two ways



Cardinal Approach

In this approach, one believes that it is measurable. One can express his or her satisfaction in cardinal numbers i.e., the quantitative numbers such as 1, 2, 3, and so on. It tells the preference of a customer in cardinal measurement. It is measured in utils.

Ordinal Approach

In this approach, one believes that it is comparable. One can express his or her satisfaction in ranking. One can compare commodities and give them certain ranks like first, second, tenth, etc. It shows the order of preference. An ordinal approach is a qualitative approach to measuring a utility.

Types of Utility

It is basically of three types

Total

The sum of the total satisfaction from the consumption of specific goods or services.

It increases as more goods are consumed.

Total Utility (T.U.) = $U_1 + U_2 + \dots + U_n$

Marginal

It is the additional satisfaction gained from each extra unit of consumption. It decreases with each additional increase in the consumption of a good.

Marginal Utility (M.U.) = $\text{Change in T.U.} / \text{Change in Total Quantity} = \Delta TU / \Delta Q$

Average

One can obtain it by dividing the total unit of consumption by the number of total units. Suppose there are total n units, then

Average Utility (A.U.) = $\text{T.U.} / \text{Number of units} = \text{T.U.} / n$

Law of Diminishing Marginal Utility: As consumer consumes more and more units of commodity the Marginal utility derived from each successive units go on declining. This is the basis of law of demand

Meaning of consumer's equilibrium

Equilibrium means a state of maximum satisfaction.

Consumer's equilibrium is a situation when a consumer spends his given income on the purchase of one or more commodities in such a way that he gets maximum satisfaction and has no urge to change this level of consumption, given the prices of commodities Conditions of equilibrium (In case of one commodity)

The consumer will be in the state of equilibrium when the following condition is fulfilled:

The marginal utility of commodity 'X' in terms of rupees is equal to the price of commodity 'X' in rupees. $[MU_x (\text{in Rs.}) = P_x (\text{Rs.})]$

i. In case of one community

$$MU_m = \frac{MU_x}{P_x} \text{ [If } MU_m = 1, MU_x = P_x \text{]}$$

(Where, MU_m = Marginal utility of money, MU_x = Marginal utility of commodity X and P_x = Price of commodity X)

Let us take the example of Fruit Ice-cream. Price of an ice-cream Scoop is Rs.30 and MU_m i.e. MU of money (Re 1) = 1 util.

Units Consumed	MU of Ice-cream Scoop (in utils) {a}	MU of Money (Re 1) (in utils) {b}	MU of Ice-Cream Scoop (in Rs.) = {a}/ {b}		Price of Ice-Cream Scoop (Rs.)
1	50	1	50	>	30
2	40	1	40	>	30
3	30	1	30	=	30
4	20	1	20	<	30
5	10	1	10	<	30

. In the given example, the level of Consumer's Equilibrium is 3 units where,

MU of Ice Cream in rupees = Price of Ice Cream in rupees i.e. Rs.30.

- Before this level i.e. at 1st and 2nd level, $MU > \text{Price}$ i.e. benefit is more than cost. So, the consumer will increase consumption to attain equilibrium.
- After this level i.e. at 4th and 5th level, $MU < \text{Price}$. i.e. benefit is less than cost. So, the consumer will cut or decrease consumption to be in the state of equilibrium. Only at the level of 3 units, the condition of consumer's equilibrium is fulfilled.

A consumer will consume that much quantity at which $MU_x = P_x$ to be in the state of equilibrium.

Consumer equilibrium (In case of two commodities)

It is a situation in which a consumer is satisfied and he has no tendency to change his pattern of consumption .In other words a consumer is in equilibrium when he derives maximum satisfaction from the goods and is in no position to rearrange his purchases

Condition :- $MU_x = MU_y = MU_m$

ii. In case of two commodity.

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y} = MU_m$$

XY and MU must be decreasing

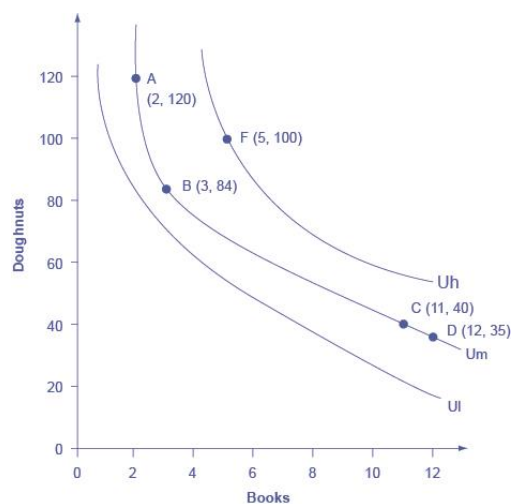
Thus, A consumer is said to be in equilibrium when he feels that he “cannot change his condition either by earning more or by spending more or by changing the quantities of thing he buys”. A rational consumer will purchase a commodity up to the point where price of the commodity is equal to the marginal utility obtained from the thing.

Indifference Curve Analysis

Economists use the vocabulary of maximizing utility to describe consumer choice. So far in the text, we have described the level of utility that a person receives in numerical terms. This section presents an alternative approach to describing personal preferences, called indifference curve analysis, which avoids the need for using numbers to measure utility. By setting aside the assumption of putting a numerical valuation on utility—an assumption that many students and economists find uncomfortably unrealistic—the indifference curve framework helps to clarify the logic of the underlying model.

What Is an Indifference Curve?

People cannot really put a numerical value on their level of satisfaction. However, they can, and do, identify what choices would give them more, or less, or the same amount of satisfaction. An indifference curve shows all combinations of goods that provide an equal level of utility or satisfaction.



For example, Figure above presents three indifference curves that represent Lilly’s preferences for the tradeoffs that she faces in her two main relaxation activities: eating doughnuts and reading

paperback books. Each indifference curve (U_l , U_m , and U_h) represents one level of utility. First we will explore the meaning of an individual indifference curve and then we will look at the relationship between different indifference curves.

Shape of an Indifference Curve

The indifference curve U_m has four points labeled on it: A, B, C, and D (see Figure 1). Since an indifference curve represents a set of choices that have the same level of utility, Lilly must receive an equal amount of utility, judged according to her personal preferences, from two books and 120 doughnuts (point A), from three books and 84 doughnuts (point B) from 11 books and 40 doughnuts (point C) or from 12 books and 35 doughnuts (point D). She would also receive the same utility from any of the unlabeled intermediate points along this indifference curve.

Indifference curves have a roughly similar shape in two ways: 1) they are downward sloping from left to right; 2) they are convex with respect to the origin. In other words, they are steeper on the left and flatter on the right. The downward slope of the indifference curve means that Lilly must trade off less of one good to get more of the other, while holding utility constant. For example, points A and B sit on the same indifference curve U_m , which means that they provide Lilly with the same level of utility. Thus, the marginal utility that Lilly would gain from, say, increasing her consumption of books from two to three must be equal to the marginal utility that she would lose if her consumption of doughnuts was cut from 120 to 84—so that her overall utility remains unchanged between points A and B. Indeed, the slope along an indifference curve is the marginal rate of substitution, which is the rate at which a person is willing to trade one good for another so that utility will remain the same.

Indifference curves like U_m are steeper on the left and flatter on the right. The reason behind this shape involves diminishing marginal utility—the notion that as a person consumes more of a good, the marginal utility from each additional unit becomes lower. Compare two different choices between points that all provide Lilly an equal amount of utility along the indifference curve U_m : the choice between A and B, and between C and D. In both choices, Lilly consumes one more book, but between A and B her consumption of doughnuts falls by 36 (from 120 to 84) and between C and D it falls by only five (from 40 to 35). The reason for this difference is that points A and C

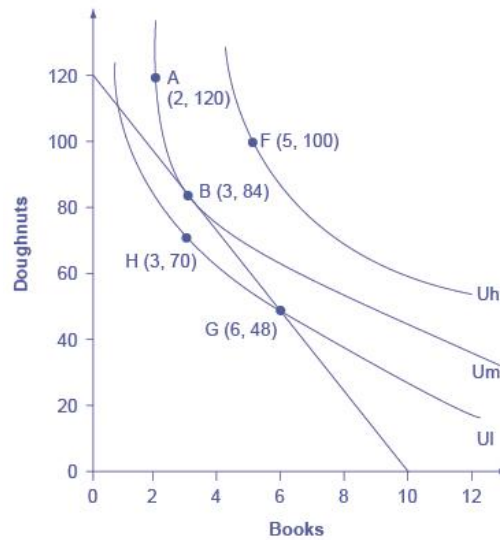
are different starting points, and thus have different implications for marginal utility. At point A, Lilly has few books and many doughnuts. Thus, her marginal utility from an extra book will be relatively high while the marginal utility of additional doughnuts is relatively low—so on the margin, it will take a relatively large number of doughnuts to offset the utility from the marginal book. At point C, however, Lilly has many books and few doughnuts. From this starting point, her marginal utility gained from extra books will be relatively low, while the marginal utility lost from additional doughnuts would be relatively high—so on the margin, it will take a relatively smaller number of doughnuts to offset the change of one marginal book. In short, the slope of the indifference curve changes because the marginal rate of substitution—that is, the quantity of one good that would be traded for the other good to keep utility constant—also changes, as a result of diminishing marginal utility of both goods.

Individuality of Indifference Curves

Each person determines his or her own preferences and utility. Thus, while indifference curves have the same general shape—they slope down, and the slope is steeper on the left and flatter on the right—the specific shape of indifference curves can be different for every person. Figure 1, for example, applies only to Lilly’s preferences. Indifference curves for other people would probably travel through different points.

People seek the highest level of utility, which means that they wish to be on the highest possible indifference curve. However, people are limited by their budget constraints, which show what tradeoffs are actually possible. The solution to this problem, i.e. the combination of goods and services that will maximize an individual’s total utility given their budget, is called the consumer equilibrium

Return to the situation of Lilly’s choice between paperback books and doughnuts. Say that books cost \$6, doughnuts are 50 cents each, and that Lilly has \$60 to spend. This information provides the basis for the budget line shown in Figure 1. Along with the budget line are shown the three indifference curves from Figure 1. What is Lilly’s utility-maximizing choice? Several possibilities are identified in the diagram.



Indifference Curves and a Budget Constraint. Lilly's preferences are shown by the indifference curves. Lilly's budget constraint, given the prices of books and doughnuts and her income, is shown by the straight line. Lilly's optimal choice will be point B, where the budget line is tangent to the indifference curve U_m . Lilly would have more utility at a point like F on the higher indifference curve U_h , but the budget line does not touch the higher indifference curve U_h at any point, so she cannot afford this choice. A choice like G is affordable to Lilly, but it lies on indifference curve U_l and thus provides less utility than choice B, which is on indifference curve U_m .

The choice of F with five books and 100 doughnuts is highly desirable, since it is on the highest indifference curve U_h of those shown in the diagram. However, it is not affordable given Lilly's budget constraint. The choice of H with three books and 70 doughnuts on indifference curve U_l is a wasteful choice, since it is inside Lilly's budget set, and as a utility-maximizer, Lilly will always prefer a choice on the budget constraint itself. Choices B and G are both on the opportunity set. However, choice G of six books and 48 doughnuts is on lower indifference curve U_l than choice B of three books and 84 doughnuts, which is on the indifference curve U_m . If Lilly were to start at choice G, and then thought about whether the marginal utility she was deriving from doughnuts and books, she would decide that some additional doughnuts and fewer books would make her happier—which would cause her to move toward her preferred choice B. Given the combination of Lilly's personal preferences, as identified by her indifference curves, and Lilly's opportunity set, which is determined by prices and income, B will be her utility-maximizing choice.

The highest achievable indifference curve touches the budget constraint at a single point of tangency. Since an infinite number of indifference curves exist, even if only a few of them are drawn on any given diagram, there will always exist one indifference curve that touches the budget line at a single point of tangency. All higher indifference curves, like U_h , will be completely above the budget line and, although the choices on that indifference curve would provide higher utility, they are not affordable given the budget set. All lower indifference curves, like U_l , will cross the budget line in two separate places. When one indifference curve crosses the budget line in two places, however, there will be another, higher, attainable indifference curve sitting above it that touches the budget line at only one point of tangency.

Budget constraints

The budget constraint is the set of all the bundles a consumer can afford given that consumer's income. We assume that the consumer has a budget – an amount of money available to spend on bundles. For now, we do not worry about where this money or income comes from, we just assume a consumer has a budget.

So what can a consumer afford? Answering this depends on the prices of the goods in question. Suppose you go to the campus store to purchase energy bars and vitamin water. If you have \$5 to spend, energy bars cost fifty cents each, and vitamin water costs \$1 a bottle, then you could buy 10 bars, and no vitamin water, no bars and 5 bottles of vitamin water, 4 bars and 2 vitamin waters and so on.

This table shows the possible combinations of energy bars and vitamin water the student can buy for exactly \$5:

Energy Bars	Bottles of Vitamin Water
10	0
8	1
6	2
4	3
2	4
0	5

It is also true that you could spend less than \$5 and have money left over. So we have to consider all possible bundles –including consuming none at all.

Note that we are focusing on bundles of two goods so that we maintain tractability (as explained in module 1), but it is simple to think beyond two goods by defining one of the goods as “money spent on everything else.”

Mathematically, the total amount the consumer spends on two goods, A and B, is:

$$P_A A + P_B B$$

where P_A is the price of good A and P_B is the price of good B. If the money the consumer has to spend on the two goods, his income, is given as I , then the budget constraint is:

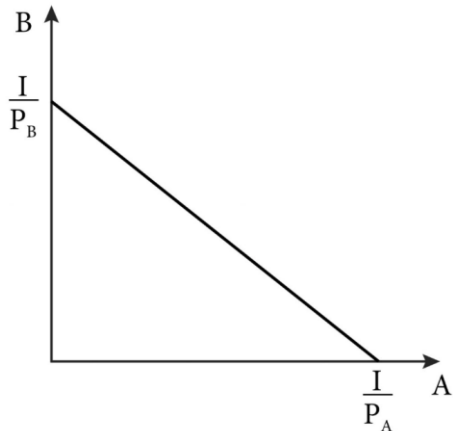
If the money the consumer has to spend on the two goods, his income, is given as I , then the budget constraint is:

$$P_A A + P_B B \leq I$$

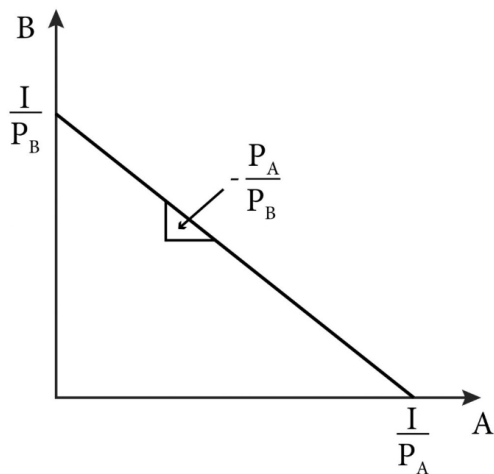
Note the inequality: This equation states that the consumer cannot spend more than his income but can spend less. We can simplify this assumption by restricting the consumer to spending all of his income on the two goods. This will allow us to focus on the frontier of the budget constraint. As we shall see in Module 4, this assumption is consistent with the more-is-better assumption – if you can consume more (if your income allows it) you should because you will make yourself better off. With this assumption in place, we can write the budget constraint as:

$$P_A A + P_B B = I$$

Graphically, we can represent this budget constraint as shown in Figure below. We call this the budget line: The line that indicates the possible bundles the consumer can buy when spending all his income.



From the graph of the budget constraint, we can see that the budget line slopes downward and has a constant slope along its entire length. This makes intuitive sense: If you buy more of one good, you are going to have to buy less of the other good. The rate at which you can trade one for the other is determined by the prices of the two goods, and they don't change.



We can find the slope of the budget line easily by rearranging equation so that we isolate B on one side. Note that in our graph, B is the good on the vertical axis, so we will rearrange our equation to look like a standard function with B as the dependent variable:

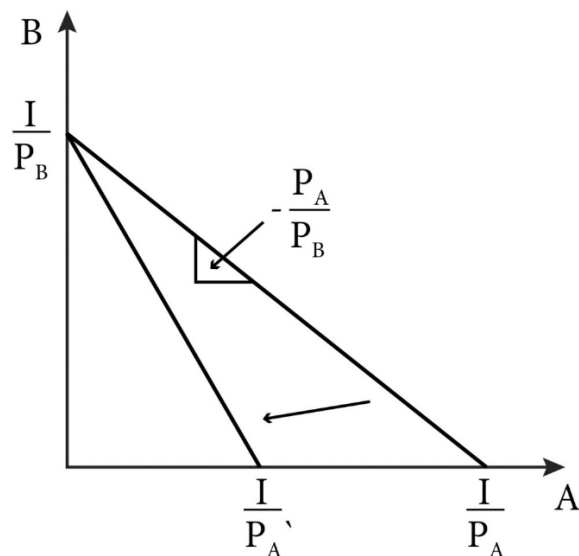
$$B = \frac{I}{P_B} - \frac{P_A}{P_B} A$$

Now, we have our budget line represented in point-slope form where:

The first part, I/P_B , is the vertical intercept.

The second part, $-P_A / P_B$, is the slope coefficient on A.

Note that the slope of the budget line is simply the ratio of the prices, also known as the price ratio. This is the rate at which you can trade one good for the other in the marketplace. To see this, let's return to the campus store with \$5 to spend on energy bars and vitamin water. Now consider an increase in the price of good A. this increase does not affect the vertical intercept, only the horizontal intercept. As P_A increases, I/P_A decreases, moving closer to the origin. This change makes the budget line 'steeper' or more negatively sloped as we can see from the slope coefficient: $-P_A / P_B$. As P_A increases, this ratio increases in absolute value, so the slope becomes more negative or steeper. What this means intuitively is that the trade-off or opportunity cost has risen. Now, the consumer has to give up more of good B to consume one more unit of good A.



Next, consider a change in income. Suppose the consumer gets an additional amount of money to spend, so I increases. I affects both intercept terms positively, so as I increases both I/P_A and I/P_B increase or move away from the origin. But I does not affect the slope: $-P_A / P_B$. Thus the shift in the budget line is a parallel shift outward – the consumer with the additional income can afford more of both.

Consumer Equilibrium

A consumer is in equilibrium when he derives maximum satisfaction from the goods and is in no position to rearrange his purchases.

Assumptions

- There is a defined indifference map showing the consumer's scale of preferences across different combinations of two goods X and Y.
- The consumer has a fixed money income and wants to spend it completely on the goods X and Y.
- The prices of the goods X and Y are fixed for the consumer.
- The goods are homogenous and divisible.
- The consumer acts rationally and maximizes his satisfaction.

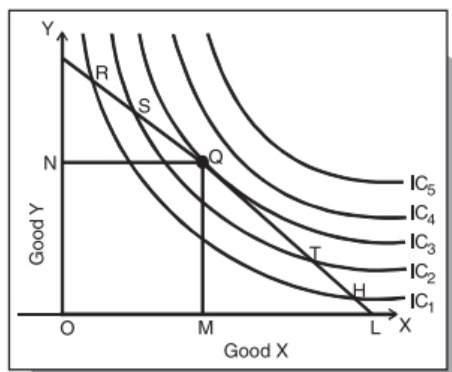
In order to display the combination of two goods X and Y, that the consumer buys to be in equilibrium, let's bring his indifference curves and budget line together.

We know that,

Indifference Map – shows the consumer's preference scale between various combinations of two goods

Budget Line – depicts various combinations that he can afford to buy with his money income and prices of both the goods.

In the following figure, we depict an indifference map with 5 indifference curves – IC₁, IC₂, IC₃, IC₄, and IC₅ along with the budget line PL for good X and good Y.



From the figure, we can see that the combinations R, S, Q, T, and H cost the same to the consumer. In order to maximize his level of satisfaction, the consumer will try to reach the highest indifference curve. Since we have assumed a budget constraint, he will be forced to remain on the budget line.

So, which combination should we choose?

Let's say that he chooses the combination R. From Fig. 1, we can see that R lies on a lower indifference curve – IC1. He can easily afford the combinations S, Q, or T which lie on the higher ICs. Even if he chooses the combination H, the argument is similar since H lies on the curve IC1 too.

Next, let's look at the combination S lying on the curve IC2. Here again, he can reach a higher level of satisfaction within his budget by choosing the combination Q lying on IC3 – higher indifference curve level. The argument is similar for the combination T since T lies on the curve IC2 too.

Therefore, we are left with the combination Q.

What happens if we chooses the combination Q?

This is the best choice since Q lies on his budget line and puts him on the highest possible indifference curve, IC3. While there are higher curves, IC4 and IC5, they are beyond his budget. Therefore, he reaches the equilibrium at point Q on curve IC3.

Notice that at this point, the budget line PL is tangential to the indifference curve IC3. Also, in this position, the consumer buys OM quantity of X and ON quantity of Y.

Since point Q is the tangent point, the slopes of line PL and curve IC3 are equal at this point. Further, the slope of the indifference curve shows a marginal rate of substitution of X for Y (MRS_{xy}) equal to $MU_x MU_y$. Also, the slope of the price line (PL) indicates the ratio between the prices of X and Y and is equal to P_x/P_y .

Understanding Consumer's Equilibrium by Indifference Curve Analysis!

Consumer equilibrium refers to a situation, in which a consumer derives maximum satisfaction, with no intention to change it and subject to given prices and his given income. The point of maximum satisfaction is achieved by studying indifference map and budget line together.

On an indifference map, higher indifference curve represents a higher level of satisfaction than any lower indifference curve. So, a consumer always tries to remain at the highest possible indifference curve, subject to his budget constraint.

Conditions of Consumer's Equilibrium:

The consumer's equilibrium under the indifference curve theory must meet the following two conditions:

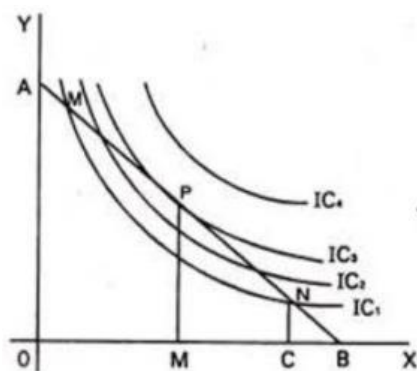
(i) $MRS_{XY} = \text{Ratio of prices or } P_X/P_Y$

Let the two goods be X and Y. The first condition for consumer's equilibrium is that

$$MRS_{XY} = P_X/P_Y$$

- If $MRS_{XY} > P_X/P_Y$, it means that the consumer is willing to pay more for X than the price prevailing in the market. As a result, the consumer buys more of X. As a result, MRS falls till it becomes equal to the ratio of prices and the equilibrium is established.
- If $MRS_{XY} < P_X/P_Y$, it means that the consumer is willing to pay less for X than the price prevailing in the market. It induces the consumer to buy less of X and more of Y. As a result, MRS rises till it becomes equal to the ratio of prices and the equilibrium is established.

Q. How does a consumer attain an equilibrium with the help of Budget line and Indifference curve?



In case of indifference curve analysis and Budget Line, the consumer attains an equilibrium at the point where the Budget line is Tangent to the highest possible IC, we can understand this concept with the help of an example above. AB is the Budget Line and IC₁ intersect it at point M and N. Both this combination will give the same level of satisfaction as they are lying on the same IC, and the amount spent on both these combinations will be same as they are lying in the same Budget Line.

IC₃ is tangent to AB at point P, which is the point of Consumer's Equilibrium. It is the combination which gives the consumer maximum level of satisfaction as it is lying in the highest IC. We can conclude that of all the combination M, N and P. The consumer will prefer the combination P as amount spent on all the combination will be same as they are lying on the same budget line. But combination P will give maximum satisfaction as it is lying on the highest Possible IC.

At the point of equilibrium, the following conditions should be fulfilled:

- 1) Budget Line should be tangent to the highest possible indifference curve.
- 2) At the point of equilibrium, the slope of Budget should be equal to the slope of IC.

The consumer will not be able to reach IC₄ due to Budget Constant. In order to attain IC₄ following conditions should be satisfied

- Price of X should fall.
- Price of Y should fall.
- Income should increase.
- Price of X and Y both should fall.

Demand forecasting:

Introduction: The information about the future is essential for both new firms and those planning to expand the scale of their production. Demand forecasting refers to an estimate of future demand for the product.

It is an ‘objective assessment of the future course of demand’. In recent times, forecasting plays an important role in business decision-making. Demand forecasting has an important influence on production planning. It is essential for a firm to produce the required quantities at the right time. It is essential to distinguish between forecasts of demand and forecasts of sales. Sales forecast is important for estimating revenue cash requirements and expenses. Demand forecasts relate to production, inventory control, timing, reliability of forecast etc. However, there is not much difference between these two terms.

Importance of Demand Forecasting:

1. **Preparing your budget:** Demand forecasting helps reduce risks and make efficient financial decisions that impact profit margins, cash flow, allocation of resources, opportunities for expansion, inventory accounting, operating costs, staffing, and overall spend. All strategic and operational plans are formulated around forecasting demand.
2. **Demand planning and scheduling production:** Demand forecasting lets you provide the products your customers want when they want them. Forecasting demand requires that order fulfillment is synced up with your marketing prior to launching. Nothing kills progress (or your reputation) faster than being sold out for weeks on end. Proper demand forecasting and inventory control can help ensure a business doesn’t buy insufficient or excessive inventory.
3. **Storing inventory:** Demand forecasting can help you spend less money on both inventory purchase orders and warehousing, as the more inventory you carry, the more expensive it is to store. Good inventory management involves having enough product on hand but not too much.
4. **Developing a pricing strategy:** Demand forecasting isn’t just about perfecting a business’s production schedule to supply demand, but it should also help price products based on the demand. By understanding the market and potential opportunities, businesses can grow, formulate competitive pricing, employ the right ecommerce marketing strategies, and invest in their growth. If you choose to slash prices or put an item on promotion, demand may temporarily increase for that product. Without that sale, you may not have

experienced the boost. If there is a limited supply of a high-demand product, you can use the scarcity principle to increase the price as an exclusive offer. You must keep an eye on new entrants though, as supply may increase.

Types of demand Forecasting:

Based on the time span and planning requirements of business firms, demand forecasting can be classified in to

1. Macro-level
 2. Micro-Level
 3. Short-term demand forecasting
 4. Long – term demand forecasting.
1. **Macro-level:** Macro-level demand forecasting looks at general economic conditions, external forces, and other broad things disrupting commerce. These factors keep a business in the know around portfolio expansion opportunities, market research intel, and other shifts in the market.
 2. **Micro-Level:** Demand forecasting at the micro-level can be specific to a particular industry, business, or customer segment (e.g., examining demand for a natural deodorant for millennial customers in Chicago, IL).
 3. **Short-term demand forecasting:** Short-term demand forecasting is limited to short periods, usually for one year. It relates to policies regarding sales, purchase, price and finances. It refers to existing production capacity of the firm. Short-term forecasting is essential for formulating a suitable price policy. If the business people expect of rise in the prices of raw materials of shortages, they may buy early. This price forecasting helps in sale policy formulation. Production may be undertaken based on expected sales and not on actual sales. Further, demand forecasting assists in financial forecasting also. Prior information about production and sales is essential to provide additional funds on reasonable terms.
 4. **Long – term forecasting:** In long-term forecasting, the businessmen should now about the long-term demand for the product. Planning of a new plant or expansion of an existing unit depends on longterm demand. Similarly, a multi-product firm must take into account the demand for different items. When forecast is mode covering long periods, the probability of error is high. It is very difficult to forecast the production, the trend of prices and the nature of competition. Hence quality and competent forecasts are essential. Prof. C. I. Savage and T.R. Small classify demand forecasting into time types.

They are

1. Economic forecasting,
2. Industry forecasting,
3. Firm level forecasting.

Economics forecasting is concerned with the economics, while industrial level forecasting is used for inter-industry comparisons and is being supplied by trade association or chamber of commerce. Firm level forecasting relates to individual firm.

Factors influencing the customer demand life cycle:

1. Seasonality
2. Competition
3. Types of goods
4. Geography

Methods of forecasting:

1. Survey Method:

Under this method, information about the desires of the consumer and opinion of exports are collected by interviewing them. Survey method can be divided into four types' viz., Option survey method; expert opinion; Delphi method and consumers interview methods.

A. Opinion survey method: This method is also known as sales-force composite method (or) collective opinion method. Under this method, the company asks its salesman to submit estimate of future sales in their respective territories. Since the forecasts of the salesmen are biased due to their optimistic or pessimistic attitude ignorance about economic developments etc. these estimates are consolidated, reviewed and adjusted by the top executives. In case of wide differences, an average is struck to make the forecasts realistic. This method is more useful and appropriate because the salesmen are more knowledge. They can be important source of information. They are cooperative. The implementation within unbiased or their basic can be corrected.

B. Expert opinion method: Apart from salesmen and consumers, distributors or outside experts may also be used for forecasting. In the United States of America, the automobile companies get sales estimates directly from their dealers. Firms in advanced countries make use of outside experts for estimating future demand. Various public and private agencies all periodic forecasts of short or long term business conditions.

C. Delphi Method: A variant of the survey method is Delphi method. It is a sophisticated method to arrive at a consensus. Under this method, a panel is selected to give suggestions to solve the problems in hand. Both internal and external experts can be the members of the panel. Panel members are kept apart from each other and express their views in an anonymous manner. There is also a coordinator who acts as an intermediary among the panelists. He prepares the questionnaire and sends it to the panelist. At the end of each round, he prepares a summary report. On the basis of the summary report the panel members have to give suggestions. This method has been used in the area of technological forecasting. It has proved more popular in forecasting. It has proved more popular in forecasting noneconomic rather than economic variables.

D. Consumers interview method: In this method the consumers are contacted personally to know about their plans and preference regarding the consumption of the product. A list of all potential buyers would be drawn and each buyer will be approached and asked how much he plans to buy the listed product in future. He would be asked the proportion in which he intends to buy. This method seems to be the most ideal method for forecasting demand.

2. Statistical Methods:

Statistical method is used for long run forecasting. In this method, statistical and mathematical techniques are used to forecast demand. This method relies on past data. The statistical method is one of the important methods of demand forecasting. Statistical methods are scientific, reliable and free from biases. The major statistical methods used for demand forecasting are:

a. Time series analysis or trend projection methods:

A well-established firm would have accumulated data. These data are analyzed to determine the nature of existing trend. Then, this trend is projected into the future and the results are used as the basis for forecast. This is called as time series analysis.

This data can be presented either in a tabular form or a graph. In the time series post data of sales are used to forecast future.

This method is useful where the organization has a sufficient amount of accumulated past data of the sales. This data is arranged chronologically to obtain a time series. Thus, the time series depicts the past trend and on the basis of it, the future market trend can be predicted. It is assumed that the past trend will continue in the future. Thus, on the basis of the predicted future trend, the demand for a product or service is forecasted.

b. Barometric Technique: Simple trend projections are not capable of forecasting turning points. Under Barometric method, present events are used to predict the directions of change in future. This is done with the help of economics and statistical indicators. Those are (1) Construction Contracts awarded for building materials (2) Personal income (3) Agricultural Income. (4) Employment (5) Gross national income (6) Industrial Production (7) Bank Deposits etc.

c. Regression and correlation method: Regression and correlation are used for forecasting demand. Based on past data the future data trend is forecasted. If the functional relationship is analyzed with the independent variable it is simple correlation. When there are several independent variables it is multiple correlation. In correlation we analyze the nature of relation between the variables while in regression; the extent of relation between the variables is analyzed. The results are expressed in mathematical form. Therefore, it is called as econometric model building. The main advantage of this method is that it provides the values of the independent variables from within the model itself.

Regression Techniques

Regression Analysis: This method establishes a relationship between the dependent variable and the independent variables. In our case, the quantity demanded is the dependent variable and income, the price of goods, the price of related goods, the price of substitute goods, etc. are independent variables. The regression equation is derived assuming the relationship to be linear. Regression Equation: $Y = a + bX$. Where Y is the forecasted demand for a product or service.

Time series forecasting

Time series forecasting is a set of methods in statistics and data science to predict some variables that develop and change over time. The underlying intention of time series forecasting is determining how target variables will change in the future by observing historical data from the time perspective, defining the patterns, and yielding short or long-term predictions on how change occurs – considering the captured patterns. Time series problems are always time-dependent and we usually look at four main components: seasonality, trends, cycles, and irregular components.

Trends. The trend component describes how the variable – Example drug sales in this case – changes over long periods of time. We see in post data that the sales revenues of antidiabetic drugs have substantially increased during the period from the 1990s to 2010s.

Seasons. The seasonal component showcases each year's wave-like changes in sales patterns. Sales were increasing and decreasing seasonally. Seasonal series can be tied to any time measurement, but these periods of time always have a fixed length and succession. We can consider monthly or quarterly patterns for sales in midsize or small e-Commerce, or track micro-interactions across a day.

Cycles. Cycles are long-term patterns that have a waveform and recurring nature similar to seasonal patterns but with variable length, they don't have a fixed time period. For example, business cycles have recognizable elements of growth, recession, and recovery. But the cycles themselves stretch in time differently for a given country throughout its history.

Irregularities. Irregular components appear due to unexpected events, like cataclysms, or are simply representative of noise in the data.